Harshit Anand

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Education

Kalinga Institute of Industrial Technology — **Aug. 2021 – Aug. 2025**

Bachelor of Technology - Computer Science and Systems Engineering [CGPA: 7.89] — Bhubaneshwar, India

B.D. Public School – 12th Boards — **2021**

CBSE

Technical Skills

- **Technologies**: Python (Ollama, LM Studio, Langchain, AssemblyAI, Seaborn, Pandas, NumPy, SKlearn, TensorFlow, PyTorch), MySQL.
- **Skills**: Machine Learning, Deep Learning, Data Analysis, Data Visualization, SQL, C/C++ and Python programming, Large Language Model, Data Cleaning and Preprocessing, Prompt Engineering, Presentation Skills.
- **Tools**: GitHub, Spreadsheet, PowerPoint, PyCharm, Visual Studio, Jupyter Notebook, Anaconda.

Experience and Projects

Audio Intelligence and Conversational Agent (AICA) (Link) — **May. 2024 – Dec. 2024**

- Developed AICA, a GPU-based system for fast transcription, summarization, and optimized audio processing efficiency.
- Built mobile platform enabling recording, uploading, managing audio with secure user login and intuitive interface.
- Used FFmpeg to convert various audio formats like MP3, WAV for unified, consistent audio processing.
- Implemented Whisper for multilingual speech recognition, real-time speaker diarization, and 95% transcription accuracy.
- Integrated AssemblyAI API for scalable English transcription, ensuring robust and reliable cross-domain performance.
- Employed Mistral-NeMo LLM for concise, accurate audio summarization, improving analysis and user accessibility.
- Automated processing workflows using LLM triggers upon new audio uploads, streamlining response and integration.

Accident Detection and Emergency Response System (Link) — **Dec. 2024 – Apr. 2025**

- Designed and deployed a sensor-based crash detection model achieving 92%+ accuracy on real-time anomaly detection.
- Implemented multiprocessing architecture handling 10+ concurrent users, improving detection latency by over 60%.
- Integrated mobile, backend, and AI modules using MongoDB for synchronized data flow and seamless communication.
- Developed real-time accident detection using smartphone accelerometer, gyroscope, speed, and GPS sensor data streams.
- Automated emergency workflow reduces manual response time by 70%, with live ambulance routing via Google Maps.
- Trained and optimized anomaly detection models for high-frequency motion data ensuring low-latency crash detection.
- Deployed full-stack emergency response system, including backend APIs and dashboards, on Render cloud hosting platform.

Deepfake Detection System (Link) — **Dec. 2023 – May 2024**

- Engineered a sophisticated model for frame-by-frame video analysis, identifying face-swapped videos with 98% accuracy.
- Significantly enhancing content authenticity and reducing manual verification workload by 50%.
- Employing ResNet50, LSTM, neural network and face recognition.
- The system captures the faces from the given input video using a face recognition module.
- Using ResNet50 model for feature extraction and LSTM model for classifying into fake or original.
- Achieved accuracy of 83% in identifying fake videos.

Real-time Face Mask Detection (Link) — **Mar. 2024 – Apr. 2024**

- Developed a computer vision model achieving 95% accuracy in detecting faces with and without masks in real-time.
- This leads to a 20% reduction in manual monitoring time.
- The dataset on which the keras sequential model has been trained has in total of 3833.
- Images of people in masks and people with no mask on stored separately, each of the categories has about 1920 images.
- Engineered an advanced machine learning model using convolutional neural networks (CNNs).
- Integrated it with webcam functionality, achieving a 95% accuracy rate in real-time object detection.

Certifications and Training

- **IBM Professional Certification** in Data Science from Coursera (Link)
- **Hacker Rank SQL** (Basic + Intermediate + Advance), **Problem Solving** (Basic + Intermediate), **Python** (Basic) (Link)
- **DevOps** certified by training and placement cell KIIT (Katallyst) (Link)